

Research Article

Quantum Behaved Particle Swarm Optimization-Based Deep Transfer Learning Model for Sugarcane Leaf Disease Detection and Classification

T. Tamilvizhi ¹, R. Surendran ², K. Anbazhagan ² and K. Rajkumar ³

¹Department of Information Technology, Vel Tech Multi Tech Dr.RangarajanDr.Sakunthala Engineering College, Chennai 600062, India

²Department of Computer Science and Engineering, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, India

³School of Computer Science and Information Technology, DMI-St.John the Baptist University, Mangochi, Malawi

Correspondence should be addressed to K. Rajkumar; rajkumar@dmisjbu.edu.mw

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Plant diseases pose a major challenge in the agricultural sector, which affects plant development and crop productivity. Sugarcane farming is a highly organized part of farming. Owing to the desirable condition for sugarcane cultivation, India stands among the second largest producers of sugarcane over the globe. At the same time, sugarcane gets easily affected by multifarious diseases which significantly influence crop productivity. The recently developed computer vision (CV) and deep learning (DL) models with an effective design can be employed for the detection and classification of diseases in sugarcane plant. The disease detection in sugarcane plant is not accurate in the existing techniques. This paper presents a quantum behaved particle swarm optimization based deep transfer learning (QBPSO-DTL) model for sugarcane leaf disease detection and classification which produces high accuracy. The proposed QBPSO-DTL method is designed and trained for the prediction of diseased leaf images. The proposed QBPSO-DTL technique encompasses the design of optimal region growing segmentation to determine the affected regions in the leaf image. In addition, the SqueezeNet model is employed as a feature extractor and the deep stacked autoencoder (DSAE) model is applied as a classification model. Finally, the hyperparameter tuning of the DSAE model is carried out by using the QBPSO algorithm. For demonstrating the enhanced outcomes of the QBPSO-DTL approach, a wide range of experiments were implemented and the results ensured the improvements of the QBPSO-DTL model.

1. Introduction

Sugarcane is a crop which is rich in sugar known as the sucrose. It can be utilized for making jaggery, white sugar, and other by-products such as bagasse and molasses. With the help of sugarcane, 75% of the sugar is produced globally. The second largest producer and the primary consumer of sugar is India. Furthermore, India is also the second largest agriculture-based industry [1]. Since sugarcane juice has a natural alkalinity its usage prevents breast cancer and prostate cancer as well. Likewise, it is advantageous for appropriate working of the liver and kidney and it is also beneficial for maintaining the blood pressure at normal

levels. However, the sugarcane plant has observed epidemics of disease that leads to crop degradation [2]. Infected sugarcane extremely infects the crop production. It is significant for monitoring the disease and health for suitable agriculture of crop. Image processing and deep learning (DL) are employed for identifying an unhealthy stem, leaf, color fruit, infected size, shape and area of leaves, etc. [3]. Sugarcane is a longer duration crop (10–16 months); it can be frequently attacked with diseases as stated in the following: the sugarcane plant is infected by different kinds of diseases resulting from bacteria, fungi, viruses, phytoplasma, or protozoans. Few sugarcane diseases are mosaic diseases such as red rot, grassy shoot, and ring spot [4].

Disease-induced crop loss differs from 10%–50% just depending on severity of the disease [5]. Hence, disease symptoms must be timely detected and appropriate measures must be instantly taken for preventing the progress or spread of the diseases. This accurate and instant diagnosis would certainly reduce the crop loss in sugar beet field [6]. Advancement in computer vision offers a possibility to enhance and expand the accuracy in plant protection practice and to increase precision agriculture [7]. The technique called image processing was introduced by various studies for the recognition and classification of sugarcane diseases. Image processing technique was utilized to extract the feature and to later identify if it is infected or not [8]. It is possible to recognize the infected areas and categorize the severity of the disease by examining the color and shape features through direct image processing method in infected leaf images. Alternatively, the disease categorization by utilizing the machine learning (ML) methods included support vector machine (SVM) and K-means clustering. Similarly, the disease categorization by utilizing DL methods included artificial neural networks (ANN) and convolutional neural networks (CNN). In recent years, the usage of DL approaches for pest and disease recognition in plants was widely studied [9]. Although several algorithms and techniques were introduced, but still there is a room for additional improvements [10].

Militante and Gerardo [11] aimed to incorporate different CNN frameworks of DL approaches for achieving the maximum accuracy rate in recognizing and detecting diseases [11]. The model was trained to detect diseases by using 14,725 images of infested sugarcane disease and healthier sugarcane leaf and the results show that it accomplished a maximal accuracy rate of 95.40%. CNN architecture consists of LeNet, VGGNet, and StridedNet which have been utilized in recognizing and detecting diseases. Padilla et al. developed a portable device that employs SVM which recognizes yellow spot disease on sugarcane leaves [12]. The study focuses on developing a model that captures and displays images of sugarcane leaf incorporated with a single unit system through image processing. The researcher trained the model for characterizing and classifying the variance among leaves that is diseased or healthier by identifying a yellow spot on the leaf. Hemalatha et al. [13] presented a DL-NN framework where the different kinds of disease that afflict the crop are forecasted by training the system on the image of an infected leaf [13]. The subsequent disease is distinguished as follows: yellow leaf disease, rust spots, cercospora leaf spot, red rot, and helmanthospura leaf spot. The method includes a CNN which is trained for image classification. Ozguven et al. developed an upgraded Fast RCNN framework by altering the parameter of the CNN, and Fast RCNN architectures for automated recognition of the leaf disease in sugar beet have been presented [14].

Thilagavathi et al. [15] focused on identifying the different diseases in a sugarcane leaf by using the image processing technique and developed a web application for the farmer to distinguish the main diseases of sugarcane [15]. The scheme gathers the image of the leaf and processes it through adoptive histogram equalization (AHE) outdated by

segmentation by means of a *k*-means clustering method. By using PCA and GLCM, the statistical characteristics, namely, skewness, variance, mean, covariance, and standard deviation, are extracted. At last, the classification and detection are performed by SVM. Quoc et al. designed a loop-mediated isothermal amplification (LAMP) for the sugarcane white leaf (SCWL) as an alternate method for the efficient and quick recognition of the SCWL phytoplasma within thirty minutes [16].

The three LAMP primer sets have been employed for detecting the 16SrXI SCWL phytoplasma. The plant cytochrome oxidase (cox) LAMP prime that amplifies a housekeeping gene from plants was utilized as the control. Shendryk et al. utilized a UAV mounted LiDAR and multispectral imaging sensor for monitoring the two sugarcane field trials with nitrogen (N) fertilization input from the wet tropic region of Australia [17]. From the six studies implemented at forty two-day intervals, we observed crop production in terms of vegetation, height, and density indices. Wang et al. developed a reverse transcription LAMP (RT-LAMP) for accurate and rapid diagnosis of SCSMV [18]. According to the conserved polyprotein gene nucleotide sequences of SCSMV isolated that differed from sorghum mosaic virus (SrMV) and sugarcane mosaic virus (SCMV), four SCSMV primers, such as P2-F3, P2-B3, P2-FIP, and P2-BIP, have been screened and designed, through a panel of sugarcane virus.

In 2021, Dhaka et al. [19] took a detailed survey on plant leaf diseases with the help of deep convolutional neural networks [19]. Plant leaf diseases are classified with some regular methods such as image rotation flipping, scaling, and cropping, translation and by adding Gaussian noise. The advantage of this work is that it provides a very detailed survey on the plant leaf diseases which is very useful for the researchers but the limitation of this work is that it is used mostly by the fundamental traditional methods and minimum DL techniques. Olusola Oluwakemi Abayomi-Alli et al. found a better solution for the cassava disease recognition [20]. It utilizes the image preprocessing through the color transformation method. The benefits of this work are that it classifies the images accurately with the help of the augmentation technique but, however, the drawback in this work is its low accuracy in detecting the diseases.

In 2021, Almadhor et al. [21] implemented the project for detecting the guava plant diseases by using the machine learning algorithms [21]. This project provides the accurate recognition results for guava plant diseases. This work needs to extend with modern techniques from DL or ML. Rehman et al. [22] developed a framework to recognize the apple leaf diseases through MASK RCNN technique [22]. Here, RCNN stands for region-based convolutional neural network. This model produces efficient results for the identification of apple leaf diseases. The issue in this work is the need to train the large number of images.

All the existing works presented in the research study focus on disease detection in sugarcane leaf with different methodologies. The presented existing works were limited in nature and vary with regions and they are not very useful in

detecting and classifying the sugarcane leaf diseases accurately. Thus, there is a need for accurate detection and classification techniques in the proposed research work. The main objective of the proposed research is better detection and classification of sugarcane leaf diseases with high accuracy. A comprehensive comparative study ensured the improvements of the QBPSO-DTL model over the other techniques in terms of different measures. Therefore, the QBPSO-DTL model was utilized as a proficient tool for leaf disease detection.

2. Materials and Methods of Novel QBPSO-DTL Approach

In this study, a novel QBPSO-DTL approach was developed for sugarcane leaf disease detection and classification. Figure 1 depicts the block diagram of QBPSO-DTL technique.

The proposed QBPSO-DTL technique is trained for the prediction of diseased leaf images, which has the ability to properly categorize the presence of sugarcane leaf disease. The proposed QBPSO-DTL technique encompasses several subprocesses, namely, preprocessing, optimal region growing segmentation, DSAE-based classification, and QBPSO-based hyperparameter tuning.

2.1. Preprocessing. The quality of the image is optimized by changing the intensity of the image for highlighting the targeted region that is infected visual part afterward data gathering is accomplished. The CLAHE approach is installed for enhancing images; it works on smaller sections of image rather than on the entire image itself. Since the name suggests, the CLAHE approach employs histogram equalisation afterward, partitioning the image into contextual regions. This makes the hidden feature of the image easily viewable via dispersal of gray value. Bilinear interpolation is used for integrating the adjoining tiles for eliminating artificially induced boundaries. The contrast inhomogeneous area is constrained for avoiding amplified noise that could be presented in the image.

2.2. Optimal Region Growing Segmentation. During the image segmentation process, the disease-affected leaf portions in the preprocessed images are correctly identified and segmented using the region growing technique. The fundamental technique that is derived from the region growth for 2 D images is given as follows:

- (i) The chosen slice of the affected leaf portions was uploaded and named as image.
- (ii) The coordinate of the initial point (pixel) of development is defined by the user.
- (iii) The color intensity of this chosen point was saved from the base value as $seed_{val}$.
- (iv) The threshold value was regarded as $thresh_{val}$ and by default is 20% gray. The threshold of total images in the following equation

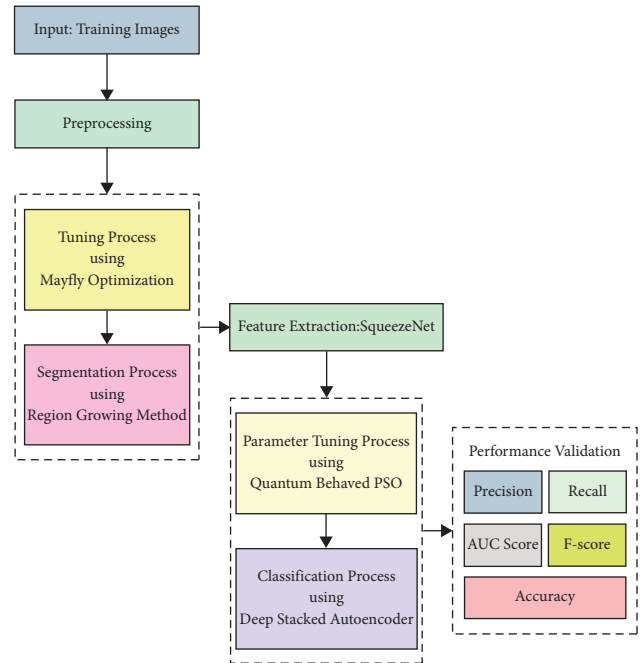


FIGURE 1: A block diagram of QBPSO-DTL technique.

$$thresh_{val} = 0.2 * graythresh(image). \quad (1)$$

- (v) The coordinate of primary pixels from an array termed as points is stored.
- (vi) The 8 pixels around the initial pixel (neighbour pixel) is regarded. It can be verified that the color intensity of these pixels is in the range of initial pixel color intensity with specific accuracy which is known as the $thresh_{val}$ value. All these points are in addition the points array offered that the subsequent criteria are met in the following equation:

$$seed_{val} = thresh_{val} \leq point_{val}, \quad (2)$$

where $thresh_{val}$ denotes the threshold value.

$seed_{val}$ indicates color intensity of the chosen point. $point_{val}$ represents the coordinate of the primary pixels from an array point.

The color intensity of chosen point ($seed_{val}$) is equal to the threshold value ($thresh_{val}$). At the same time, the threshold value ($thresh_{val}$) is less than or equal to the coordinate of primary pixels from an array point ($point_{val}$). It is called “queuing.” The above criteria are again tested to novel neighbour pixel that is in the queue at preceding step. Based on this procedure, remaining pixels are not eligible to enqueue in the queue. At this point, every pixel from the point array is mentioned as the affected leaf portion. Furthermore, the outermost pixels are displayed as curved affected leaf portion boundaries. For optimal selection of the seed value, the MFA is applied. The MFA is a swarm intelligence-based algorithm which is inspired by the mating and flight characteristics of mayflies (MFs). The major stages involved in the MFA are the movement of male MFs (MMF)

- (1) Initialize control parameter, problem dimension, and limits
- (2) Create MMF population y_i ($i = 1, 2, \dots, M$) and the velocity v_{mi}
- (3) Create FMF population x_i ($i = 1, 2, \dots, N$) and its velocities v_{fi}
- (4) Determine solution by the use of a fixed objective function.
- (5) Compute gbest and pbest
- (6) while (termination condition)
- (7) Upgrade velocity and location of the MMFs and FMFs based on the velocity and location bounds
- (8) Determine solution by the use of a fixed objective function.
- (9) Sort MFs population
- (10) Mate mayflies population
- (11) Validate off springs
- (12) Segregate offspring to male and female arbitrarily
- (13) Substitute worst solutions with the best new solutions
- (14) Upgrade pbest and gbest
- (15) end while
- (16) Store optimum solution

ALGORITHM 1: Pseudo-code of mayfly algorithm.

and female MF (FMF), crossover, and mutation. The place of the MMFs can be defined by the following equation:

$$x_i^{t+1} = x_i^t + v_i^{t+1}, \quad (3)$$

where x_i^t and x_i^{t+1} denote the present and newly obtained locations correspondingly. v_{ij}^{t+1} depicts the velocity as defined in the following equation:

$$v_{ij}^{t+1} = v_{ij}^t + a_1 \times e^{-\beta r_p^2} \times (pbest_{ij} - x_{ij}^t) + a_2 \times e^{-\beta r_g^2} \times (gbest_j - x_{ij}^t), \quad (4)$$

where a_1 and a_2 denote the positive social and cognitive constant. x_{ij}^t indicates particle location in dimensions j when v_{ij}^t demonstrates the velocity. In addition, pbest and gbest demonstrate the global best solutions. $pbest_{ij}$ can be represented the following equation:

$$pbest_j = \begin{cases} x_i^{t+1}, & \text{iff } (x_i^{t+1}) < f(pbest_i), \\ \text{issavedsame}, & \text{otherwise.} \end{cases} \quad (5)$$

gbest indicates the best solution of pbest. r_p and r_g denote Cartesian coordinates and β implies predefined visibility coefficient. The mayfly algorithm was proposed with a better hybridization of the PSO algorithm [23,24].

The nuptial dance is an essential scene for optimal MF and varied the velocity as provided in the following equation:

$$v_{ij}^t = v_{ij}^t + d \times r, \quad (6)$$

where d and r denote coefficients of nuptial dance and arbitrary value. The place of the FMFs can be defined using the following equation:

$$y_i^{t+1} = y_i^t + v_i^{t+1}, \quad (7)$$

where y_i^t and y_i^{t+1} imply the present and newly attained locations, and v_i^{t+1} denotes velocity, as given in the following equation:

$$v_{ij}^{t+1} = \begin{cases} v_{ij}^t + a_2 \times e^{-\beta r_{mf}^2} (x_{ij}^t - y_{ij}^t), & \text{iff } (y_i) > f(x_i), \\ v_{ij}^t + fl \times r, & \text{iff } (y_i) \leq f(x_i), \end{cases} \quad (8)$$

where v_{ij}^t represents the velocity of female particle i in dimension j . a_2, β , r_{mf} , and fl are the positive attraction values, fixed visibility constant, Cartesian distance, and random walk value, respectively.

The mating process is defined by the use of crossover and mutation, which can be represented using the following equation:

$$\text{Crossover} \begin{cases} \text{offspring1} = L \times \text{male} + (1 - L) \times \text{female}, \\ \text{offspring2} = L \times \text{female} + (1 - L) \times \text{male}, \end{cases} \quad (9)$$

$$\text{Mutation} \{ \text{offspring}'_n = \text{offspring}_n + \text{randvalue},$$

where L indicates the arbitrary value.

2.3. SqueezeNet-Based Feature Extraction. At the time of feature extraction, the segmented outcomes are fed into the SqueezeNet model to produce the feature vectors. Stanford and Berkeley in 2017 presented a light weight SqueezeNet [25]. The light weight CNN redesigned the network framework according to the present CNN model for achieving the aim of minimizing the computational complexity and the amount of parameters. The basic component of SqueezeNet is the fire module that comprises of the expand and the squeeze layers. The squeeze layers use a 1×1 convolutional layer to convolve the input feature [26]. The major goal is to minimize the amount of channels of input feature, viz., reduction dimension. The expand layers use 1×1 and 3×3 convolutional operations, correspondingly, and later concatenate the convolutional outcomes. The integration of this layer could efficiently minimize the number of parameters. The major concept of SqueezeNet is to interconnect the fire module in a cascaded format. Figure 2 illustrates the infrastructure of the SqueezeNet. It reduces

the amount of parameters and gives full play to the features of the fire module in the network.

Specifically, the pooling layer is positioned in the back position. The aim is to offer a large activation map for the convolution layer. Since the activation map retains numerous data, the larger the activation map of the convolution layer, the higher the classification performance of the network. It must be notable that ReLU activation function is utilized in squeezing and expanding the layers of the SqueezeNet. After the fire module, the dropout technique has been utilized, and the output specification of the dropout depends on the dimension of the convolution layer. Additionally, for compressing the network of SqueezeNet, we rejected the FC layer. Lastly, the softmax classifiers were employed to output the classification result, and the output specification is based on the classification.

SqueezeNet initiates with an independent convolution layer (conv1) and cascade 8 fire modules, such as fire2-fire9, while ending the cascade with a convolution layer (conv10). At last, the global average pooling layer is utilized rather than the FC layer for output. From the start to the end, the networks gradually increase the amount of convolutional layer in fire modules. After the fire4, fire8, and conv1 layers, the max pooling with step size of 2 is employed, correspondingly, and the global average pooling is implemented after conv10.

2.4. Image Classification Using DSAE Model. Finally, the DSAE model receives the useful feature vectors and then classifies sugar plant leaf diseases. AE is a type of unsupervised 3-layer NN that has 2 parts of encoding and decoding, comprising input, hidden, and output layers [27]. The function has been formulated in the following equation:

$$\begin{aligned} y &= f(x) \\ &= s(W^{(1)}x + b^{(1)}), \end{aligned} \quad (10)$$

where $\chi \in R^{d \times 1}$ represents the input vector, d stands for the dimensional of input data, $y \in R^{r \times 1}$, r refers the count of hidden layer units, $W^{(1)} \in R^{r \times d}$ defines the input weight to hidden layer, and $b^{(1)} \in R^{r \times 1}$ signifies the input bias to hidden layer. simplifies the activation function that is generally nonlinear. The generally utilized activation function is the sigmoid function $s(x) = 1/(1 + e^{-x})$ and tanh function $s(x) = e^x - e^{-x}/e^x + e^{-x}$. The role of the decoded is for mapping the formulation y of the hidden layer back to original input. The function has been formulated in the following equation:

$$\begin{aligned} x &= g(y) \\ &= s(W^{(2)}y + b^{(2)}), \end{aligned} \quad (11)$$

where $W^{(2)} \in R^{d \times r}$, $b^{(2)} \in R^{d \times 1}$. Therefore, the reconstruction error to all data is shown in the following equation:

$$L = \|x - g(f(x))\|^2. \quad (12)$$

The determined cost function is shown in the following equation:

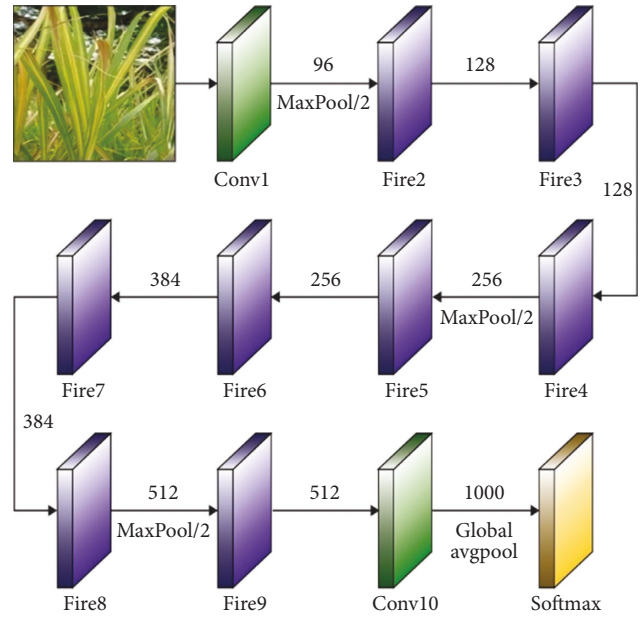


FIGURE 2: SqueezeNet architecture.

$$\begin{aligned} J(W, b) &= \left[\frac{1}{N} \sum_{i=1}^N \left(\frac{1}{2} \|x^{(i)} - g(f(x^{(i)}))\|^2 \right) \right] \\ &+ \frac{\lambda}{2} \sum_{l=1}^2 \sum_{i=1}^{S_l} \sum_{j=1}^{S_{l+1}} (W_{ji}^{(l)})^2, \end{aligned} \quad (13)$$

where $x^{(i)}$ refers to the i^{th} instance, $W_{ji}^{(l)}$ is the connection weighted against the i^{th} unit of l^{th} layer and j^{th} unit of $(l+1)^{th}$ layer, N refers to the amount of instances, and S_l stands for the amount of units from the l^{th} layer. An optimum solution W and b of the model is attained by the error BP and the batch gradient descent technique.

Various AEs are “stacked” in a greedy layer-by-layer method to initialize the weight of DNN for attaining a deep SAE. An SAE architecture has two AEs organized in a cascade manner and assists to reduce the data dimension by selecting the proper feature. Next, this feature is fed to the softmax classification for classifying problems. It is a possibility-based linear classification that estimates the likelihood distribution of inputs over n distinct inputs. It employs the significant feature and assists in increasing the classifier accuracy.

$$F(x_j) = \frac{\text{Exp}(x_i)}{\sum_{j=0}^k \text{Exp}(x_j)}, \quad \text{where } i = 0, 1, \dots, k. \quad (14)$$

Equation (14) signifies the softmax function. It evaluates the exponential of input x_i and the summation of exponential of each input value and the fraction of this results from the resultant of softmax function.

2.5. Parameter Tuning Using QBPSO Model. For optimal parameter tuning of the SAE model, the QBPSO algorithm has been utilized. Based on the principle of quantum

mechanics, QPSO algorithm has been developed. By using DELTA potential well, the PSO is employed for the quantum space. The quantum space particle utilized wave function in the following equation:

$$|\Psi|^2 dx dy dz = Q dx dy dz. \quad (15)$$

Amongst them, $|\Psi|^2$ denotes the square of module of the wave function, and it represents the probability density of a particle in a position to appear. Q indicates the likelihood density function and satisfies the normalization criteria in the following equation:

$$\int_{-\infty}^{+\infty} |\Psi|^2 dx dy dz = \int_{-\infty}^{+\infty} Q dx dy dz = 11. \quad (16)$$

Consider that the D dimension of (for the dimension of variable related to with) quantum space has population that comprises of n particles. The location of the i th particle is $X_i = (x_{i1}, x_{i2}, \dots, x_{iD})$, and the particle through the history of the optimal position is $P_i = (p_{i1}, p_{i2}, \dots, p_{iD})$; afterward, each particle of the optimal location is $P_g = (p_{g1}, p_{g2}, \dots, p_{gD})$. In quantum space, the position of particle after the particle gets through the stochastic simulation of Monte Carlo measurement is shown in the following equation:

$$x_{id} = p_{id} \pm \frac{L}{2} \ln \left(\frac{1}{u} \right) (i = 1, 2, \dots, n) (d = 1, 2, \dots, D). \quad (17)$$

Amongst them, u denotes the arbitrary value within $[0, 1]$. L is attained by the particle existing location and past location is $L = 2 \cdot \beta \cdot |p_{id} - x_{id}|$ in the following equation:

$$x_{id}(t+1) = p_{id} \pm \beta \cdot |p_{id} - x_{id}(t)| \cdot \ln \left(\frac{1}{u} \right). \quad (18)$$

Amongst them, t indicates the iteration value. β shows the contraction expansion factor and the variable of quantum PSO. For avoiding the premature convergence, Rout et al. enhanced the quantum PSO [28], presenting mbest in the model and in the following equation:

$$\begin{aligned} \text{mbest}(t) &= \frac{1}{n} \sum_{i=1}^n p_i(t) \\ &= \left[\frac{1}{n} \sum_{i=1}^n p_{i1}(t), \frac{1}{n} \sum_{i=1}^n p_{i2}(t), \dots, \frac{1}{n} \sum_{i=1}^n p_{iD}(t) \right], \end{aligned} \quad (19)$$

whereas p_i denotes the optimal location of i th particles and n shows the amount of particles. mbest finds the average optimal position of n particles and resolves problem-based dimension in the following equation:

$$L = 2 \cdot \beta \cdot |\text{mbest} - x_{id}|, \quad (20)$$

$$x_{id} = p_{id} \pm \beta \cdot |\text{mbest}_d - x_{id}| \cdot \ln \left(\frac{1}{u} \right).$$

Thus, quantum PSO particle update equation is determined and can be described in the following equation:

$$\begin{aligned} p_{id} &= \phi \cdot p_{id} + (1 - \phi) \cdot p_{gd}, \\ \text{mbest}(t) &= \frac{1}{n} \sum_{i=1}^n p_i(t), \\ &= \left[\frac{1}{n} \sum_{i=1}^n p_{i1}(t), \frac{1}{n} \sum_{i=1}^n p_{i2}(t), \dots, \frac{1}{n} \sum_{i=1}^n p_{iD}(t) \right], \text{mbest}(t) = \frac{1}{n} \sum_{i=1}^n p_i(t), \\ &= \left[\frac{1}{n} \sum_{i=1}^n p_{i1}(t), \frac{1}{n} \sum_{i=1}^n p_{i2}(t), \dots, \frac{1}{n} \sum_{i=1}^n p_{iD}(t) \right], \\ X_i(t+1) &= p_i \pm \beta \cdot |\text{mbest} - X_i(t)| \cdot \ln \left(\frac{1}{u} \right). \end{aligned} \quad (21)$$

Amongst them, ϕ denotes the arbitrary value within $[0, 1]$; other variables are similar to the one mentioned above. The QBPSO approach resolves a FF for attaining enhanced classifier performance. It defined a positive integer for representing an optimum performance of the candidate solution [29]. In this case, the minimizing of the classifier error rate is regarded as FF and is provided in equation (22).

An optimum solution has minimal error rate and a worst solution gains an enhanced error rate [30–32].

$$\begin{aligned} \text{fitness}(x_i) &= \text{classifier error rate}(x_i) \\ &= \frac{\text{number of misclassified images}}{\text{total number of images}} * 100. \end{aligned} \quad (22)$$



FIGURE 3: Sample images.

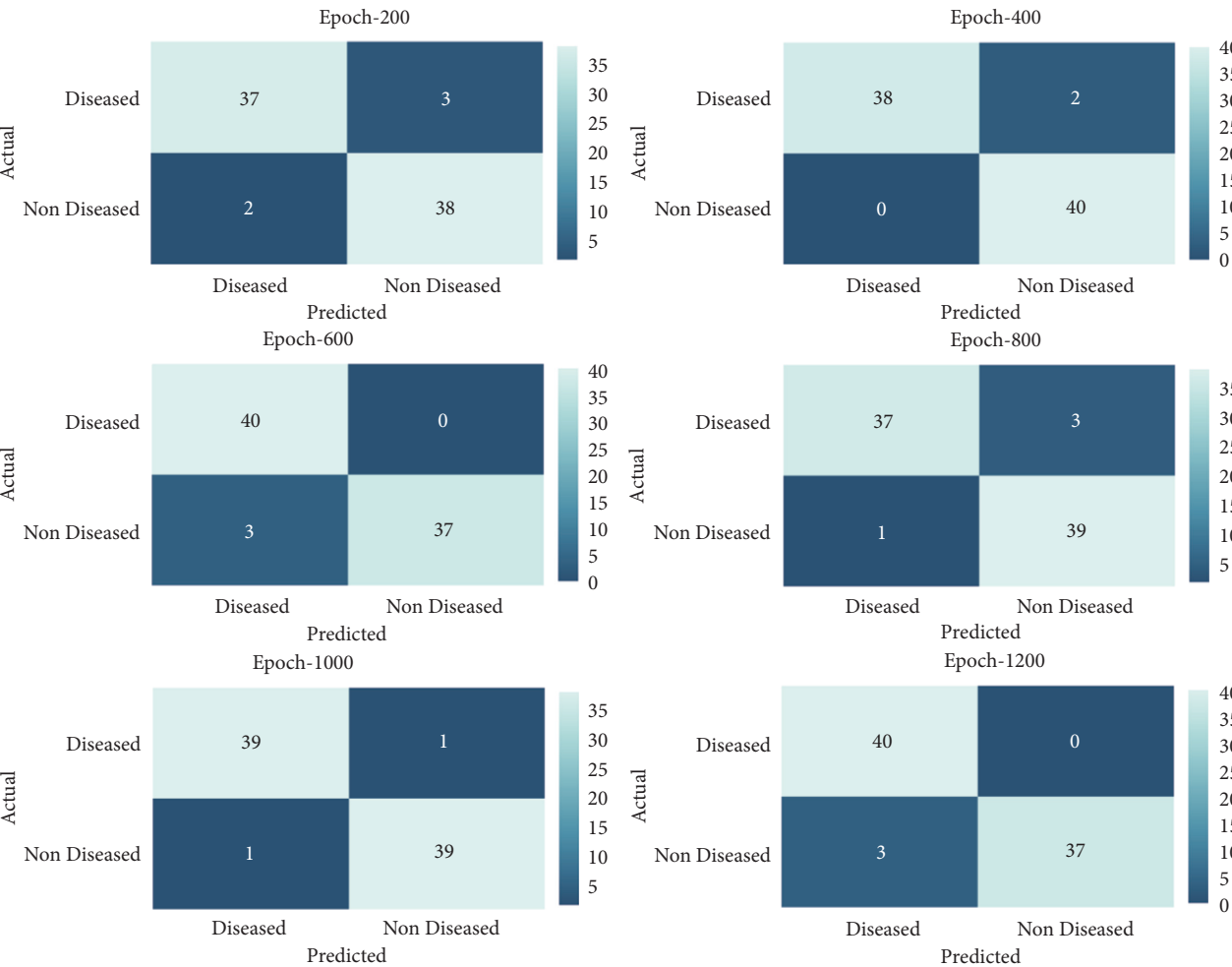


FIGURE 4: Confusion matrix of QBPSO-DTL technique under various epochs.

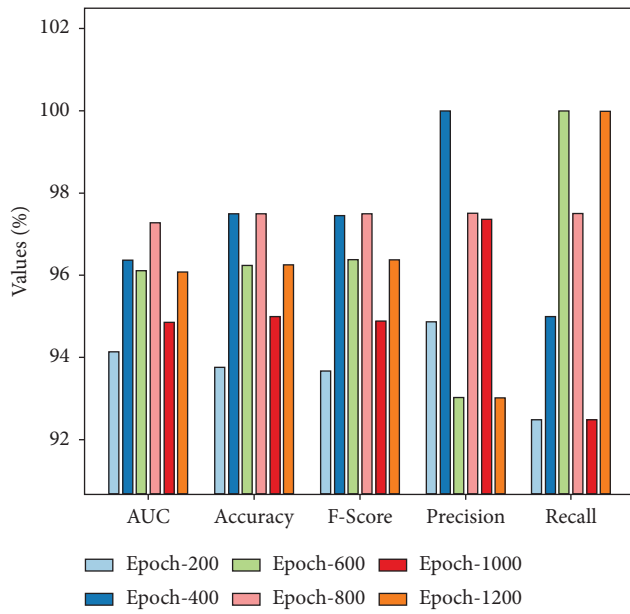


FIGURE 5: Result analysis of QBPSO-DTL technique under different epochs.

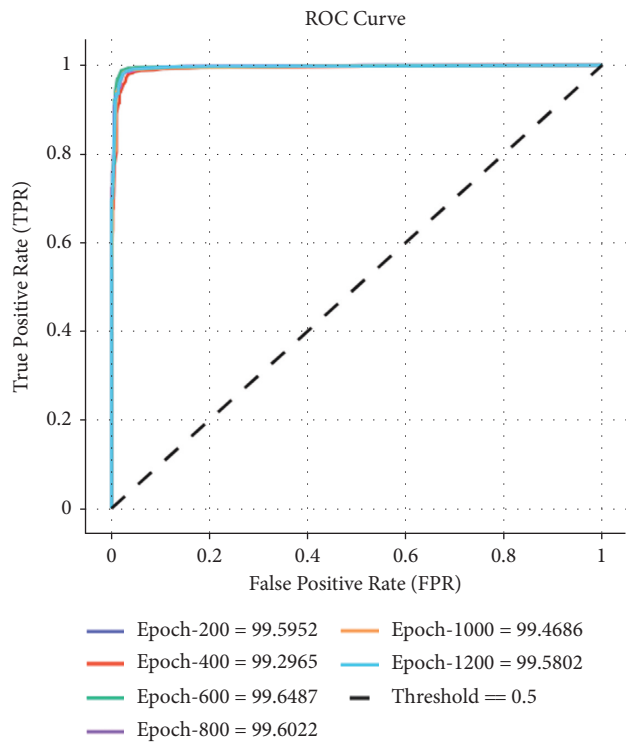


FIGURE 6: ROC analysis of QBPSO-DTL technique under different epochs.

3. Results and Discussion

In this section, the sugarcane leaf disease detection analysis of the QBPSO-DTL model takes place using its own dataset. It comprises two subdatasets, namely, training and testing datasets with two classes, namely, diseased and nondiseased images. The SqueezeNet is used as the deep learning

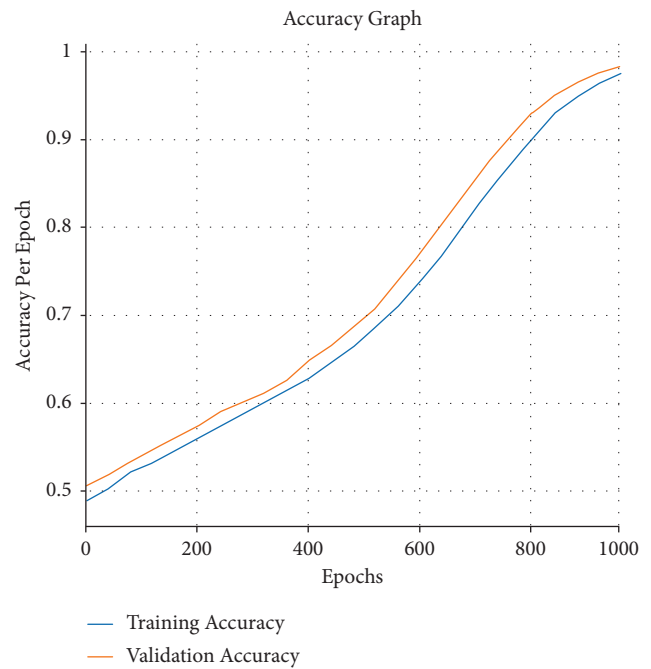


FIGURE 7: Accuracy analysis of QBPSO-DTL technique.

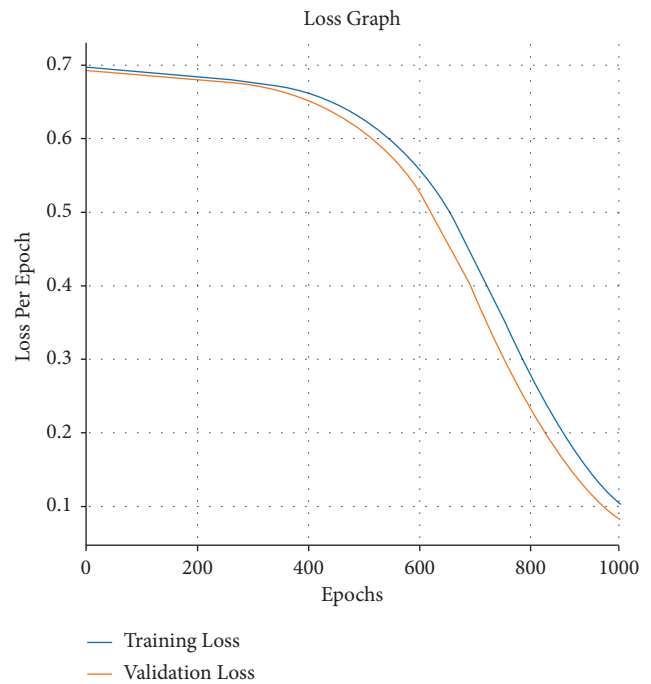


FIGURE 8: Loss analysis of QBPSO-DTL technique.

architecture, and the transfer learning is used as the training mechanism, 80 : 20 is the percentage of training and testing ratio. A few sample images of sugarcane leaves with diseased and nondiseased classes are illustrated in Figure 3. The training dataset holds 80 images under each diseased and nondiseased class. Similarly, the testing dataset includes 40 images under diseased and nondiseased classes.

Even though better outcomes were described in the DL models, the variety of the datasets used is constrained.

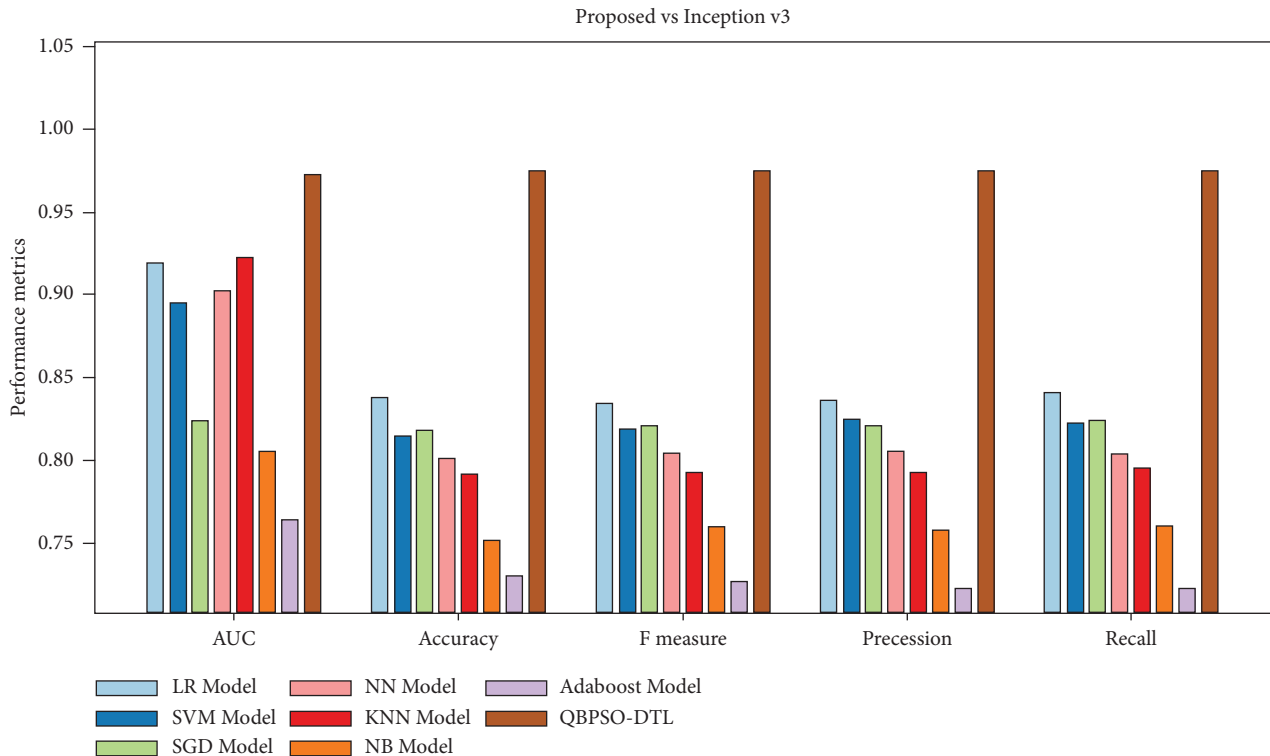


FIGURE 9: Comparative analysis of QBPSO-DTL technique with different Inception v3.

A larger dataset (comprised of hundreds of images) is needed for the CNN training. Unfortunately, for recognizing plant leaf diseases, these diverse and large datasets have not been gathered. Now, transfer learning (TL) is the efficient method for training the strength of CNN classifier for recognizing plant leaf diseases [33–35]. The TL method allows the adaptation of pretrained CNN by retraining them with small dataset and its dispersal is distinct from the larger datasets formerly utilized for training the network from scratch. In fact, it is efficient that utilizes CNN model pretrained on the ImageNet dataset and later retrains for recognizing leaf diseases. Thus, the integration of DL and TL offers a novel method to resolve the issue of constrained datasets of plant diseases. This paper presents a quantum behaved particle swarm optimization-based deep transfer learning (QBPSO-DTL) model for sugarcane leaf disease detection and classification [36,37].

Figure 4 shows that the QBPSO-DTL model has the capability of identifying images into two classes, namely, diseased and nondiseased. For instance, with 200 epochs, the QBPSO-DTL model has identified 37 images under diseased class and 38 images under nondiseased class. In addition, with 600 epochs, the QBPSO-DTL method has identified 40 images under diseased class and 37 images under nondiseased class. Along with that, with 800 epochs, the QBPSO-DTL system has identified 37 images under diseased class and 39 images under nondiseased class. At the same time, with 1200 epochs, the QBPSO-DTL algorithm has identified 40 images under diseased class and 37 images under nondiseased class. Figure 4 highlights the set of

confusion matrices created by the QBPSO-DTL model on the classification of sugarcane leaf disease images under the various epochs.

Figure 5 exhibits the overall sugarcane leaf disease detection outcomes of the QBPSO-DTL model under various epochs. The results indicated that the QBPSO-DTL model has resulted in an enhanced classification outcome under every epoch. For instance, with 200 epochs, the QBPSO-DTL model has offered AUC of AUC, acc_y , $F_{measure}$, $prec_n$, and $recal$ of 94.12%, 93.75%, 96.67%, 94.87%, and 92.50%, respectively. Along with that, with 600 epochs, the QBPSO-DTL methodology has obtainable AUC of AUC, acc_y , $F_{measure}$, $prec_n$, and $recal$ of 96.12%, 96.25%, 96.39%, 93.02%, and 100%, correspondingly. Moreover, with 800 epochs, the QBPSO-DTL methodology has presented AUC of AUC, acc_y , $F_{measure}$, $prec_n$, and $recal$ of 97.28%, 97.50%, 97.50%, 97.50%, and 97.50%. Eventually, with 1000 epochs, the QBPSO-DTL model has accessible AUC of AUC, acc_y , $F_{measure}$, $prec_n$, and $recal$ of 94.86%, 95%, 94.87%, 97.37%, and 92.50%, respectively. Meanwhile, with 1200 epochs, the QBPSO-DTL algorithm has offered AUC of AUC, acc_y , $F_{measure}$, $prec_n$, and $recal$ of 96.08%, 96.25%, 96.39%, 93.02%, and 100%, correspondingly.

Figure 6 illustrates the ROC analysis of the QBPSO-DTL approach under distinct epochs. The figure revealed that the QBPSO-DTL algorithm has reached improved outcomes with the maximum ROC of 99.5952 under epoch of 200.

The results demonstrated that the QBPSO-DTL method has accomplished higher validation accuracy in relation to training accuracy. It can also be noticed that the accuracy values get saturated with the count of epochs. The accuracy

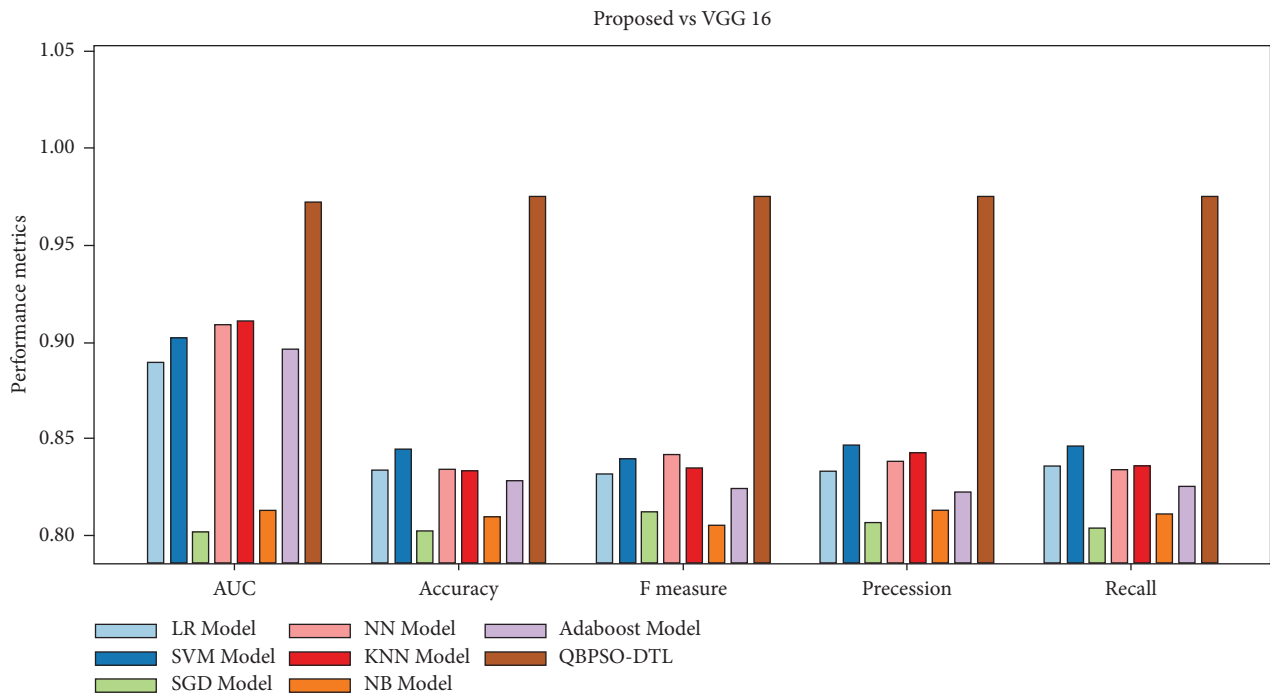


FIGURE 10: Comparative analysis of QBPSO-DTL technique with different VGG 16.

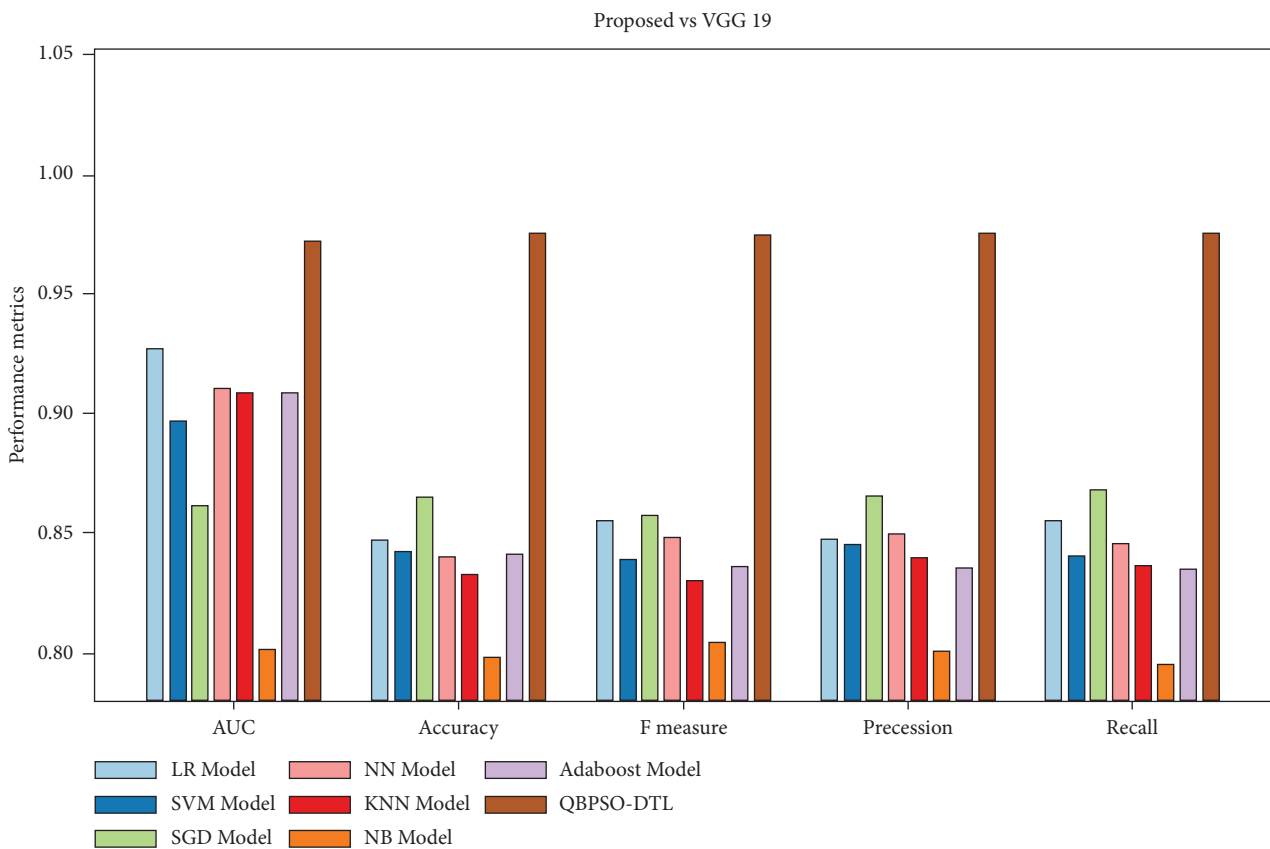


FIGURE 11: Comparative analysis of QBPSO-DTL technique with different VGG 19.

outcome analysis of the QBPSO-DTL approach on test data is showcased in Figure 7.

The loss outcome analysis of the QBPSO-DTL method on test data is exhibited in Figure 8. The figure demonstrates that the QBPSO-DTL system has denoted the reduced validation loss over the training loss. It can be additionally noticed that the loss values get saturated with count of epochs.

Figure 9 offers a detailed comparative result analysis of the QBPSO-DTL model with different classifiers of the Inception v3 model. The results indicated that the Inception v3-KNN, Inception v3-NB, and Inception v3-Adaboost models have shown ineffective outcomes with the minimal values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$ 38,39,40, followed by the Inception v3-NN model which has obtained slightly enhanced values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. In line with, the Inception v3-SVM and Inception v3-SGD models have resulted in reasonable values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. However, the presented QBPSO-DTL method has outperformed the other approaches with maximum AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$ of 0.9728, 0.9750, 0.9750, 0.9750, and 0.9750.

Figure 10 offers a detailed comparative result analysis of the QBPSO-DTL technique with distinct classifiers of VGG v16 model. The outcomes indicated that the VGG v16-KNN, VGG v16-NB, and VGG v16-Adaboost models have shown ineffective outcomes with the minimal values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. Then, the VGG v16-NN approach has obtained somewhat higher value of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. Likewise, the VGG v16-SVM and VGG v16-SGD systems have resulted in reasonable values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. At last, the presented QBPSO-DTL technique has demonstrated the other algorithms with maximum AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$ of 0.9728, 0.9750, 0.9750, 0.9750, and 0.9750.

Figure 11 offers a detailed comparative result analysis of the QBPSO-DTL approach with different classifiers of VGG v19 method. The results stated that the VGG v19-KNN, VGG v19-NB, and VGG v19-Adaboost models have demonstrated ineffective outcomes with the minimal values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. Moreover, the VGG v19-NN model has obtained slightly maximal values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. Furthermore, the VGG v19-SVM and VGG v19-SGD systems have resulted in reasonable values of AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$. Eventually, the presented QBPSO-DTL methodology has portrayed the other approaches with maximal AUC, $accu_y$, $F_{measure}$, $prec_n$, and $reca_l$ of 0.9728, 0.9750, 0.9750, 0.9750, and 0.9750.

After observing the aforementioned results and discussion, it can be ensured that the QBPSO-DTL system has the ability to outperform the other methods in terms of distinct measures.

4. Conclusion

In this study, a new QBPSO-DTL approach was developed for sugarcane leaf disease detection and classification. The proposed QBPSO-DTL technique is trained for the

prediction of diseased leaf images, which has the ability to properly categorize the presence of sugarcane leaf disease. The proposed QBPSO-DTL technique encompasses several subprocesses, namely, preprocessing, optimal region growing segmentation, DSAE-based classification, and QBPSO-based hyperparameter tuning. The performance validation of the QBPSO-DTL model is carried out and the results are assessed under several aspects. A comprehensive comparative study ensured the improvements of the QBPSO-DTL model over the other techniques in terms of different measures. Therefore, the QBPSO-DTL model was utilized as a proficient tool for leaf disease detection and classification. In future, the deep instance segmentation models can be developed to improve the accuracy of detection and classification.

Data Availability

The datasets generated and/or analysed during the current study are not publicly available due to the extension of the submitted research work but are available from the corresponding author on reasonable request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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